Ch 6. Gambling and Data compression

"Gambling in terms of information theory!"

↳ Horse Racing.

Note. $\sum_{i=1}^{N} b_i = 1$

$b_i \ge 0, \ \forall i.$

Horses | Winning rate | pay off (= Return per 1, if won) | Gambler's investment (betting)

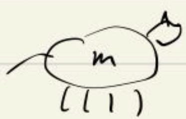
p_1 not Known | O_1 Known. | $b_1 \longrightarrow (b_1, p_1, O_1)$

$\vdots$ | $\vdots$ | $\vdots$ | $\vdots$

* Notation

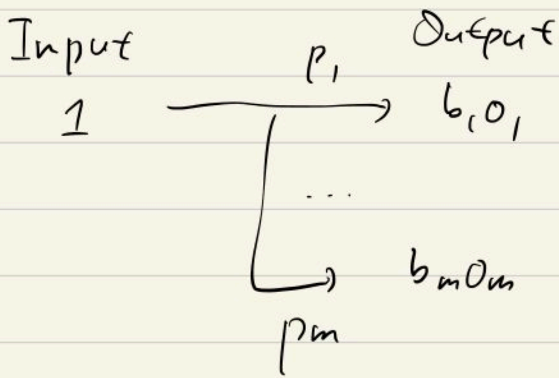
$(b_1, \cdots, b_n) = \mathbf{b}$

$(p_1, \cdots, p_n) = \mathbf{p}$.



p_m | O_m | b_m

Given | Can be Chosen



Utility″

(Low risk wanted.)

@″ Maximum $(\underbrace{\text{Output}}_{\mathbb{E}}?)$

$$\rightarrow \mathbb{E} = \sum_{i=1}^{m} p_i b_i o_i$$

$$\le \left(\sum_{i=1}^{m} b_i\right) \max_i (p_i o_i) = \max_i (p_i o_i).$$ ✓

* Convex function might be used:

$$\sum_i p_i f(x_i) \searrow f\left(\sum_i p_i x_i\right) \quad \text{as} \quad \max \|x_i - x_j\| \to 0$$

↓

f is convex $\}$ $f = \log$

Def″ (Doubling rate)

$$W(b;p) = \mathbb{E}[\log S(X)] = \sum_{x=1}^{m} p_x \log(b_x o_x)$$

Notation
X_i : i.i.d r.v. with $\mathbb{P}(X_i = b_j o_j) = p_j$, $\forall i$.

$S(X_i) := b(X_i) o(X_i) \longrightarrow$ Return at i^{th} race

$S_n(X_i) := \prod_{i=1}^{n} S(X_i)$ = Wealth after n races.

($\rightarrow$ Underlying assumption ; p_i & o_i is not change)

Observation

$$\frac{1}{n} \log S_n = \frac{1}{n} \sum_{i=1}^{n} \log S(X_i) \xrightarrow{LLN} \mathbb{E}[\log S(X)] \text{ in probability}$$

$$\Rightarrow S_n \doteq 2^\wedge(n \underbrace{\mathbb{E}(\log(S(X))}_{= W(b;p)})$$

$\longrightarrow$ $W > 0$; profit
$W = 0$; same money
$W < 0$; lose money.

Thm 6.1.1. $S_n \doteq 2^{nW(b;p)}$

Prop. $W^*(p) := \max_b W(b,p)$ is achieved if $b = p$

We want to

$$\begin{cases} \text{Maximize } \overbrace{\sum p_i \ln (b_i o_i)}^{=f} & \xrightarrow[\text{Lagrange multiplier method}]{} \quad \nabla_b f \,/\!/\, \nabla_b g \\ \text{subject to } \overbrace{\sum b_i - 1 = 0}^{=g} & \\ \quad (0 \leq b_i \leq 1) & \end{cases}$$

$\underline{o_i > 0}$

$$\Longleftrightarrow \left(\frac{p_1}{b_1}, \cdots \frac{p_n}{b_n}\right) /\!/ (1, 1, \cdots 1)$$

$$\longrightarrow \frac{p_1}{b_1} = \frac{p_2}{b_2} = \cdots = \frac{p_n}{b_n} = \frac{p_1 + p_2 + \cdots + p_n}{b_1 + b_2 + \cdots + b_n} = 1$$

$$\Longrightarrow p_i = b_i \quad \forall i \Longrightarrow p = b \qquad \square$$

Thm 6.1.2.

$$W^*(p) = \sum_i p_i \log o_i - H(p)$$

pf. $\displaystyle \sum_i p_i \log (b_i o_i)\Big|_{b=p} = \sum_i p_i \log o_i - \underbrace{\sum_i p_i \log \frac{1}{p_i}}_{H(p).}$

ex 6.1.1. Two horses $(h_1, p_1, 2)$, $(h_2, p_2, 2) \longrightarrow$ [2 for 1]

$$W^*(p) = \sum_i p_i \underbrace{\log O_i}_{=1}\Big|_{O_i=2} - H(p) = 1 - H(p) \Longrightarrow S_n \doteq 2^{n(1-H(p))}$$

* Low Entropy $\longrightarrow$ High return.

Recall. Let p, q be two pmf's. Then, relative entropy, or Kullback-Leibler distance (KL distance) btw p and q is

$$D(p\|q) = \sum_x p(x) \log \frac{p(x)}{q(x)}.$$

KL distance is not a mathematical distance ($\because$ Triangle inequality, Symmetry $\rightarrow$ False!)
but we have

$$D(p\|q) \geq 0, \text{ with equality iff } p \equiv q.$$

* Bookmaker's estimate (Reference estimate)

(=Bookie, 마권업자(배당률 정하는 사람))

$$
\begin{array}{ccccc}
 & & & \text{Bookie} \longrightarrow & \text{Payoff} \\
h_1 & p_1 & \overset{?}{=} & r_1 & \frac{1}{r_1} \\
h_2 & p_2 & \overset{?}{=} & r_2 & \frac{1}{r_2} \\
\vdots & \vdots & & \vdots & \\
h_m & p_m & \overset{?}{=} & r_n & \frac{1}{r_n}
\end{array}
$$

* Assume $\sum_i \frac{1}{o_i} = 1$, set $r_i := \frac{1}{o_i}$, $r = (r_1, r_2, \cdots, r_n)$.

Then we may regard r as a pmf, and

$$W(b,p) = \sum_i p_i \log(b_i o_i) = \sum_i p_i \log\left(\frac{b_i}{p_i}\frac{p_i}{r_i}\right)$$

$$= D(p\|r) - D(p\|b).$$

→ Heuristic: (Distance of bookie's estimate from true distribution)

— (Distance of gambler's estimate from true distribution) > 0

Gambler's desire

i.e. gambler have to make 'more accurate estimate'

(=smaller $D(p\|\cdot)$) than bookie's, to make a profit.

Thm 6.1.?. (Conservation Thm)

Given $(h_1, p_1, m), \cdots, (h_m, p_m, m)$ (so-called m-for-1 odd),

$$W^*(p) + H(p) = \log m$$

pf. $W^*(p) = \sum_i p_i \log o_i \Big|_{o_i = m} - H(p) = \log m - H(p)$. $\square$

6.2. Gambling & Side information.

- What if we have information 'Y' relevant to the outcome?

$$W^*(x) = \max_{\substack{b \\ b(x) \geq 0 \\ \sum_x b(x) = 1}} \sum_x p(x) \log b(x) o(x)$$

$$\Big\downarrow \text{'Y'}$$

$$W^*(X|Y) := \max_{\substack{b(x|y) \\ b(x,y) \geq 0 \\ \sum b(x|y) = 1}} \sum_{x,y} p(x,y) \log (b(x|y) o(x))$$

$p(x,y)$ = prob that x^{th} horse win, when y is observed,

$b(x,y)$ = bets on x, when y is observed,

$$H(X|Y) = -\sum_{x,y} p(x,y) \log p(x|y)$$

Recall (Mutual information)

$$I(X;Y) := \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)} = H(X) - H(X|Y)$$

$$= H(Y) - H(Y|X) \geq 0$$

Heuristic : Reduction in the uncertainty of X due to knowledge of Y.

Y X.

Thm 6.2.1.

$$\Delta W := W^*(X|Y) - W^*(X) \underset{\text{Thm}}{=} I(X;Y)$$

pf. $W(b, X|Y)$ is maximal if $b(x|y) = p(x|y)$.

$$W^*(X|Y) = \sum p(x,y) \log (b(x|y)\, o(x)) \Big|_{b(x|y) = p(x|y)}$$

$$\left(\sum_y p(x,y) = p(x)\right) \leftarrow = \sum_x p(x) \log o(x) + \sum_{x,y} p(x,y) \log p(x|y)$$

$$= W^* + H(X) - H(X|Y) = W^* + I(X;Y). \qquad \square$$

6.3. Dependent Horse Races and entropy

X_i : not i.i.d $\longrightarrow$ Assuption ; $\{X_n\}$ froms a stochastic process,

X_n depends on $X_1, \ldots, X_{n-1}$.

Ex ① (Red & Black)

Setup

2 horses

- 52 trump cards : 26 red, 26 black.
- Bets : next color.
- 2-for-1 : Return $\begin{cases} \text{twice} & \text{if bets on right color} \\ \text{nothing} & \text{else} \end{cases}$
- Discard the used card.

Ex ①, (Red & Black)

Setup,

- 52 trump cards : 26 red, 26 black. (→ horses)
- Bets : next color.
- 2-for-1 : Return $\begin{cases} \text{twice} & \text{if bets on right color} \\ \text{nothing} & \text{else} \end{cases}$
- Discard the used card.

(red, black)

$$\left(\frac{1}{2}, \frac{1}{2}\right) \xrightarrow{\text{red}} \left(\frac{25}{51}, \frac{26}{51}\right)$$

$$\left(\frac{1}{2}, \frac{1}{2}\right) \xrightarrow{\text{black}} \left(\frac{26}{51}, \frac{25}{51}\right)$$

Ex ②: (Guessing the alphabet)

appl ⑦ :

[Gambling & Data Compression]

In summary, "good gambler $\longrightarrow$ good data compressor!".

Setup: $X_1, \cdots, X_n$: seq. of binary r.v.s we want to compress.

Bets are given by

$$b(x_{k+1} \mid x_1, \cdots, x_k) \geq 0, \quad \sum_{x_{k+1}} b(x_{k+1} \mid x_1, \cdots, x_k) = 1. \qquad 2\text{-for-}1.$$

$$\Longrightarrow S_n = 2^n \prod_{k=1}^{n} b(x_k \mid x_1, \cdots, x_{k-1}) = 2^n b(x_1, \cdots, x_n).$$

fixed, known to compressor & decompressor

Consider data compression

$$x = x_1 x_2 \cdots x_n \in \{0,1\}^{n} \longmapsto c = c_1 c_2 \cdots c_k \in \{0,1\}^k$$

- Equip lexicographical order on x,

$$\left(x_1 x_2 \cdots x_n < y_1 y_2 \cdots y_n \iff {}^\exists k \text{ s.t. } x_i = y_i \ \ i = 1, 2, \cdots, k-1 \ \ \& \ \ x_k < y_k \right)$$

① Calculate $F(x(n)) := \sum_{x'(n) \leq x(n)} 2^{-n} S_n(x'(n)) \in [0,1]$

② For $\underline{k} = \lceil n - \log S_n(x(n)) \rceil$, define the compressed code by

$$\frac{\lfloor 2^n F(x(n)) \rfloor}{2^n} = .c_1 c_2 \cdots c_k \longmapsto c := (c_1, \cdots, c_k).$$

Binary expression

③ Find $x(n)$ s.t.

$$\sum_{x'(n) < x(n)} 2^{-n} S(x'(n)) \notin \left[.c_1 c_2 \cdots c_k, \ .c_1 c_2 \cdots c_k + \left(\tfrac{1}{2}\right)^k \right], \text{ but}$$

$$\sum_{x'(n) \leq x(n)} 2^{-n} S(x'(n)) \in \underline{\left[.c_1 c_2 \cdots c_k, \ .c_1 c_2 \cdots c_k + \left(\tfrac{1}{2}\right)^k \right]}.$$

Observation : $x(n)$ is decompression of c.

④ We can recover x from C_j^k

$\therefore$ n-bit code can be compressed by k-bits code,

$k = \lceil n - \log S(x(n)) \rceil \Longrightarrow n - k = \lfloor \log S(x(n)) \rfloor$ bits saved!

Good at gambling $\longrightarrow$ high S

$\longrightarrow$ More bits saved $\longrightarrow$ Good compression.